

DNS Sinkhole Lab — Threat Detection with Pi-hole & Wazuh SIEM

Overview

This lab demonstrates the deployment of a DNS sinkhole using Pi-hole integrated with Wazuh SIEM in a virtualized home lab environment. The objective was to intercept and log DNS queries to known malicious and tracking domains, redirect them to a null address (0.0.0.0), and forward those events to a SIEM for centralized visibility.

This simulates a real-world defensive security control used by SOC teams and network defenders to detect compromised hosts, block C2 communication, and generate threat intelligence from DNS telemetry.

Lab Environment

Component	Details
Hypervisor	Proxmox VE 9.1.1
DNS Sinkhole	Pi-hole v6.4.1 (installed on Ubuntu 22.04)
SIEM	Wazuh v4.7.5
Client VM	Windows 11 (WIN11-CLIENT)
Network	Internal lab network — 192.168.1.0/24
Pi-hole / Wazuh IP	192.168.1.233
Client IP	192.168.1.189

Tools Used

- **Pi-hole** — Network-wide DNS sinkhole and query logger

- **Wazuh SIEM** — Log ingestion, alerting, and security event management
 - **Proxmox** — Type 1 hypervisor hosting all VMs
 - **StevenBlack Hosts List** — Consolidated malware and tracking domain blocklist (92,277 domains)
 - **nslookup** — DNS query tool used to simulate and verify sinkholing
 - **nano** — Text editor used to configure Wazuh ossec.conf
-

Objectives

- Deploy Pi-hole as a DNS sinkhole on an existing Ubuntu VM
 - Load a threat intelligence blocklist containing known malicious domains
 - Configure a Windows 11 client to use Pi-hole as its DNS resolver
 - Simulate DNS queries to blocked domains and confirm sinkholing
 - Forward Pi-hole logs to Wazuh for centralized SIEM monitoring
 - Verify end-to-end detection chain from client query to SIEM ingestion
-

MITRE ATT&CK Mapping

Technique	ID	Description
Command and Control — Application Layer Protocol: DNS	T1071.004	Adversaries use DNS to communicate with C2 infrastructure. The sinkhole intercepts these queries before they reach the attacker.
Exfiltration Over Alternative Protocol	T1048	DNS-based exfiltration attempts are intercepted and logged by Pi-hole.
Indicator Removal — DNS Cache Poisoning (Defense)	T1565	Sinkholing redirects malicious DNS responses to null, preventing resolution of attacker-controlled domains.
Network Traffic Content Inspection (Detection)	T1040	Pi-hole logs provide full DNS query visibility, enabling detection of beaconing or suspicious lookup patterns.

Phase 1 — Pi-hole Installation

Pi-hole was installed directly on the Wazuh Ubuntu VM (192.168.1.233) using the official installer script.

Steps:

1. Updated system packages: `sudo apt update && sudo apt upgrade -y`
2. Resolved interrupted dpkg state: `sudo dpkg --configure -a`
3. Ran Pi-hole installer: `curl -sSL https://install.pi-hole.net | bash`
4. Selected **ens18** as the network interface
5. Selected **Cloudflare (DNSSEC)** as the upstream DNS provider
6. Enabled query logging — **Show everything** privacy mode
7. Installation completed successfully

Result: Pi-hole installed and active at <http://192.168.1.233/admin> with 92,277 default blocklist domains loaded.

Phase 2 — Blocklist Configuration

A consolidated threat intelligence blocklist was added to Pi-hole to expand coverage beyond the default list.

Blocklist added:

```
https://raw.githubusercontent.com/StevenBlack/hosts/master/hosts
```

Gravity update run to load blocklist:

```
sudo pihole -g
```

Result: 92,277 unique domains confirmed in gravity database. Blocklist status: Enabled.

Phase 3 — Client DNS Configuration

The Windows 11 client VM (192.168.1.189) was configured to use Pi-hole as its primary DNS resolver.

Settings applied:

- Preferred DNS: 192.168.1.233
- Alternate DNS: 8.8.8.8

Verification: DNS resolver confirmed as `pi.hole` via `nslookup`.

Phase 4 — Sinkhole Verification

A DNS query was made from the Windows 11 client to a known blocked domain to confirm the sinkhole was functioning.

Command run on Win11 client:

```
nslookup doubleclick.net
```

Result:

```
Server:  pi.hole
Address: 192.168.1.233

Name:     doubleclick.net
Addresses: ::
          0.0.0.0
```

The domain resolved to `0.0.0.0` — confirming the sinkhole intercepted the query and returned a null address instead of the real IP. The Pi-hole query log confirmed the block with client IP 192.168.1.189 recorded.

Phase 5 — Wazuh Log Integration

Wazuh was configured to ingest Pi-hole DNS logs for centralized SIEM monitoring.

Configuration added to `/var/ossec/etc/ossec.conf` :

```
<localfile>
  <log_format>syslog</log_format>
```

```
<location>/var/log/pihole/pihole.log</location>
</localfile>
```

Wazuh manager restarted:

```
sudo systemctl restart wazuh-manager
```

Pi-hole log verified streaming:

```
sudo tail -f /var/log/pihole/pihole.log
```

Result: Wazuh successfully ingesting Pi-hole dnsmasq log entries. DNS queries from the Win11 client visible in real-time log stream including cached, forwarded, and blocked domain lookups.

Findings

- DNS sinkhole successfully intercepted queries to known malicious and tracking domains
 - 92,277 domains blocked across ad, malware, and tracking categories
 - Windows 11 client DNS traffic fully visible through Pi-hole — including Microsoft telemetry domains, tracking domains, and the simulated doubleclick.net query
 - Pi-hole query log confirmed blocked queries with client IP, timestamp, domain, and action
 - Wazuh configured to monitor Pi-hole log file, providing SIEM-level visibility into DNS activity across the lab network
-

Screenshots

Screenshot	Description
screenshots/01-wazuh-dashboard.png	Wazuh dashboard showing 2 active agents
screenshots/02-pihole-install-complete.png	Pi-hole installation complete screen
screenshots/03-pihole-dashboard-	Pi-hole dashboard — Active, 92,277 domains

baseline.png	
screenshots/04-blocklist-added.png	StevenBlack blocklist added and Enabled
screenshots/05-gravity-update.png	Gravity update terminal — 92,277 domains loaded
screenshots/06-win11-dns-settings.png	Win11 DNS configured to 192.168.1.233
screenshots/07-nslookup-sinkholed.png	nslookup doubleclick.net returning 0.0.0.0
screenshots/08-pihole-dashboard-active.png	Pi-hole dashboard showing 2 blocked queries, 6.3% blocked
screenshots/09-pihole-query-log.png	Query log showing blocked domains from Win11 client
screenshots/10-wazuh-ossec-config.png	ossec.conf with Pi-hole localfile block added
screenshots/11-pihole-log-streaming.png	Live Pi-hole log stream in terminal

Key Takeaways

- DNS sinkholes are a low-cost, high-visibility defensive control that can be deployed quickly in any network environment
- Integrating Pi-hole with a SIEM like Wazuh creates a full detection pipeline from DNS query to security alert
- Even a default blocklist catches significant telemetry and tracking traffic — demonstrating the volume of outbound DNS activity on a standard Windows endpoint
- This technique directly maps to real SOC workflows where DNS logs are a primary source of threat hunting data

Author

Akel Swinton — IT Systems Administrator | Cybersecurity

github.com/akel-swinton